# Environment Management Portal

**Tools & Automation**

- Provisioning Env
- Generate Helm Repository
- Generate Pipeline ( CI/CD Process )
- Provisioning Tools / Service

**Goals:**

- Reduce dependency on devops engineers / eliminate devops engineer it self.

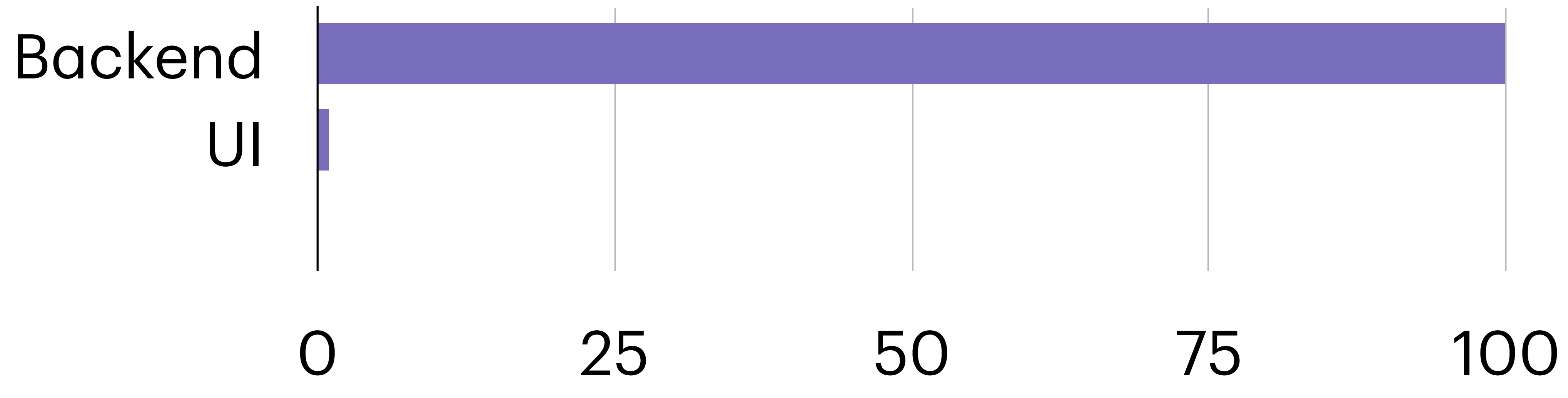
# Provisioning Env

## Tools & Automation

**Feature**:

- Automate deploy loadbalancer L4
- Automate deploy kubernetes cluster ( onpremise )
- Automate deploy kubernetes single host ( onpremise )
- Automate deploy data store ( storage class )
- One Click Install

**Development Progress:**




**How powerful are these tools?**

- They can create a Kubernetes cluster in approximately 30 minutes ( FYI 1 cluster = 7 vm minimum) & depending on the internet connection.
- Reduce the dependencies on Devops engineers for env setup

**Another Target:**

- Dashboard UI for tools ( end of year )


```
{
    "platformService": "k8s",
    "data": {
        "infoServerDst": [
            {
                "ipAddress": "10.64.20.197",
                "sudoUser": "root",
                "role": "master",
                "label": "master-0"
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            {
                "ipAddress": "10.64.20.198",
                "sudoUser": "root",
                "role": "master",
                "label": "master-1"
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            {
                "ipAddress": "10.64.20.199",
                "sudoUser": "root",
                "role": "master",
                "label": "master-2"
            },
            {
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                "sudoUser": "root",
                "role": "worker",
                "label": "worker-1"
            },
            {
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                "sudoUser": "root",
                "role": "worker",
                "label": "worker-2"
            }
        ],
```

# Use Case in Project:

- WAD has 7 regions that require Kubernetes clusters. Each cluster consists of 7 vms, so we need to manage a total of 49 Vms.
- A senior engineer needs approximately 20 - 30 minutes to configure 1 vm.
- With these tools, it takes approximately 30 minutes to install 1 cluster ( 7 vms ).

**Comparation:**

| | Minutes | Capacity | Total VM | Total hour |
|---|---|---|---|---|
| **Engineer** | 20 | 1 | 49 | 16.33333333 |
| **Tools** | 20 | 7 | 49 | 2.333333333 |

**Summary:**

It is **14** hours faster than doing it manually / 87,5% more faster.